Supplementary Material

The Ranking Transfers, the Threshold Does Not: Early At-Risk Prediction in Unseen Courses

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The numbers in this document are produced by the same script as those of the article (`src/08_paper_assets.py`), from the outputs of the code repository. Unless stated otherwise, the cut-off is at 25% of the presentation, only students still enrolled at the cut-off are included, protocol (B) is the strict leave-one-module-out, and means are taken over the seven modules.

S1 Remedies per module

Table S1 gives, module by module, the comparison summarized in Table 4 of the article, in the scenario where a random sample of $k = 100$ labelled students is available.

Table S1: Remedies per module (25% cut-off, random sample of $k = 100$ labelled students of the target module for Platt and the intercept-only update, mean of 50 draws, standard deviation across draws in parentheses; all rules are evaluated on the remaining students). Top: flag-rate drift Δ^{ref} against the absolute-feature model with the module seen. Bottom: precision. Rules as in Table V of the article.

Module	Raw	Platt	Intercept	Top-q	Top-q (EM)	Relative
Flag-rate drift Δ^{ref}						
AAA	+0.10	+0.07 (0.05)	+0.07 (0.05)	+0.07	−0.05	+0.02
BBB	+0.19	+0.06 (0.04)	+0.06 (0.04)	+0.05	+0.37	+0.03
CCC	−0.03	+0.03 (0.03)	+0.03 (0.03)	−0.02	−0.05	+0.02
DDD	−0.05	+0.02 (0.03)	+0.01 (0.03)	−0.04	−0.11	−0.05
EEE	+0.05	+0.10 (0.04)	+0.08 (0.04)	+0.15	−0.01	+0.03
FFF	−0.03	+0.07 (0.04)	+0.07 (0.04)	+0.05	−0.16	+0.01
GGG	+0.51	−0.05 (0.04)	−0.03 (0.04)	−0.03	+0.56	−0.14
Mean $\lvert\Delta^{\text{ref}}\rvert$	0.14	0.06	0.05	0.06	0.19	0.04
Mean |flagged − risk|	0.17	0.06	0.06	0.07	0.22	0.06
Precision						
AAA	0.38	0.39	0.38	0.38	0.42	0.45
BBB	0.54	0.60	0.60	0.61	0.47	0.67
CCC	0.78	0.74	0.74	0.77	0.69	0.72
DDD	0.79	0.75	0.76	0.78	0.81	0.81
EEE	0.70	0.66	0.67	0.60	0.74	0.72
FFF	0.76	0.69	0.70	0.70	0.87	0.76
GGG	0.38	0.55	0.54	0.56	0.37	0.64
Mean	0.62	0.62	0.63	0.63	0.62	0.68

S2 Cohort size

Table S2 shows how the rank model performs when ranks are computed within random sub-cohorts of the target module, over 20 random partitions. With about 50 students or more, ranks are as reliable as when they are computed on the whole presentation. Below 20 students, the gain falls off sharply.

Table S2: Ranks computed in random sub-cohorts of n students of the target module (25% cut-off, protocol (B), means over modules and over 20 random partitions; interval: 2.5–97.5% of the module mean across partitions). Shift: absolute change of the share flagged relative to ranks computed on the whole presentation.

n	AUC (B)	95% interval	Shift	|flagged − risk|	Precision	F1
10	0.778	[0.773, 0.781]	0.042	0.082	0.677	0.617
20	0.794	[0.791, 0.797]	0.016	0.065	0.675	0.641
50	0.802	[0.800, 0.804]	0.006	0.059	0.676	0.651
100	0.805	[0.804, 0.806]	0.004	0.058	0.678	0.655
200	0.806	[0.805, 0.807]	0.003	0.058	0.678	0.656
whole presentation	0.808	—	0.000	0.057	0.680	0.658

S3 Average precision

Average precision, the area under the precision–recall curve, leads to the same conclusions as the AUC (Table S3). This also holds for the rarer "Withdrawn only" target, whose prevalence is 0.137.

Table S3: Average precision (area under the precision–recall curve) at the 25% cut-off, means over modules.

Target	Model	Features	Prevalence	AP (A)	AP (B)
W only	forest	absolute	0.137	0.231	0.212
W only	forest	relative	0.137	0.250	0.230
W or F	boosting	absolute	0.387	0.748	0.728
W or F	boosting	relative	0.387	0.770	0.762
W or F	forest	absolute	0.387	0.736	0.720
W or F	forest	relative	0.387	0.764	0.755
W or F	gru	absolute	0.387	0.735	0.729
W or F	gru	relative	0.387	0.765	0.763

S4 Intersecting groups

Table S4 gives the false-positive rate of protocol (B) when gender is crossed with disability and with the most and least deprived IMD quintiles. With absolute features, disadvantages accumulate: the rate is 0.496 for women in the most deprived quintile and 0.169 for men in the least deprived one. With relative features, the remaining gaps follow deprivation rather than gender. Women and men in Q1 have rates of 0.260 and 0.234, against 0.129 for men in Q5.

Table S4: False-positive rates of protocol (B) for intersecting groups (25% cut-off), with the number of students not at risk in each group and 95% cluster-bootstrap intervals.

Features	Crossing	Group	n (negatives)	FPR [95% CI]
absolute	gender x disability	female × no disability	6,460	0.397 [0.386, 0.409]
absolute	gender x disability	female × disability	668	0.490 [0.450, 0.527]
absolute	gender x disability	male × no disability	7,716	0.205 [0.195, 0.214]
absolute	gender x disability	male × disability	538	0.297 [0.258, 0.335]
absolute	gender x IMD	female × Q1	1,327	0.496 [0.471, 0.525]
absolute	gender x IMD	female × Q5	1,223	0.346 [0.320, 0.373]
absolute	gender x IMD	male × Q1	1,194	0.290 [0.265, 0.315]
absolute	gender x IMD	male × Q5	1,729	0.169 [0.152, 0.187]
relative	gender x disability	female × no disability	6,460	0.183 [0.174, 0.193]
relative	gender x disability	female × disability	668	0.231 [0.200, 0.263]
relative	gender x disability	male × no disability	7,716	0.168 [0.159, 0.176]
relative	gender x disability	male × disability	538	0.240 [0.207, 0.276]
relative	gender x IMD	female × Q1	1,327	0.260 [0.238, 0.282]
relative	gender x IMD	female × Q5	1,223	0.134 [0.115, 0.155]
relative	gender x IMD	male × Q1	1,194	0.234 [0.208, 0.259]
relative	gender x IMD	male × Q5	1,729	0.129 [0.114, 0.145]

Table S5: Context descriptors associated with the three transfer axes (Spearman's ρ, Benjamini–Hochberg $q < 0.10$ within level, 69 tests in total). Mod.: modules; Pres.: presentations, which are not independent (they share the module-level LOMO model). Difficulty: AUC (A); Rank. cost: AUC (A) − AUC (B); Drift: flag-rate drift. Exploratory associations.

Level (n)	Axis	Descriptor	ρ	q
Mod. (7)	Difficulty	Share aged 35+	−0.96	0.016
Pres. (22)	Difficulty	Share aged 35+	−0.79	0.000
Pres. (22)	Difficulty	VLE activity types	+0.54	0.037
Pres. (22)	Difficulty	At-risk rate	+0.49	0.060
Pres. (22)	Rank. cost	Assessments due before cut-off	+0.57	0.030
Pres. (22)	Rank. cost	First deadline	−0.45	0.097
Pres. (22)	Drift	Withdrawn share among at-risk	−0.68	0.008
Pres. (22)	Drift	Share aged 35+	+0.64	0.015
Pres. (22)	Drift	At-risk rate	−0.61	0.023
Pres. (22)	Drift	Median clicks	−0.59	0.028
Pres. (22)	Drift	Share without A-level	+0.56	0.033
Pres. (22)	Drift	Assessment-structure distance	+0.52	0.048
Pres. (22)	Drift	VLE activity types	−0.51	0.048
Pres. (22)	Drift	Assessments due before cut-off	−0.44	0.099

Table S6: Model families and temporal information at the 25% cut-off (students still enrolled; means over modules). Forest and Boosting: aggregated features; Boosting + seq.: aggregated features plus weekly clicks and active days over the complete weeks before the cut-off and trend features; GRU: recurrent network on the weekly sequence plus all static features. Cost: AUC (A) − AUC (B); Drift: mean absolute flag-rate drift; GGG: drift in module GGG.

Features	Model	AUC (A)	AUC (B)	Cost	Drift	GGG
Absolute	Forest	0.800	0.787	+0.014	0.135	+0.51
	Boosting	0.805	0.787	+0.018	0.137	+0.53
	Boosting + seq.	0.808	0.791	+0.017	0.131	+0.52
	GRU	0.799	0.792	+0.007	0.127	+0.47
Relative	Forest	0.816	0.808	+0.008	0.021	+0.00
	Boosting	0.818	0.811	+0.007	0.051	+0.01
	Boosting + seq.	0.819	0.812	+0.007	0.054	+0.02
	GRU	0.817	0.815	+0.003	0.015	+0.00

S5 Context descriptors associated with transfer

Table S5 lists the associations between context descriptors and the three transfer quantities that remain after Benjamini–Hochberg control. With seven modules, and presentations that are not independent of one another, they should be read as exploratory.

S6 Other learners and sequences

Gradient boosting and a GRU network trained on weekly activity sequences show the same drift with absolute features, and the same correction with relative features (Table S6, Fig. S1). Temporal information adds little at this early stage. On unseen modules, the GRU does better than the forest by only 0.006 to 0.008 in AUC, depending on the cut-off.

S7 Target definition

Once the population is restricted to students still enrolled at the cut-off, the "Withdrawn only" target amounts to predicting future withdrawal, which is rarer and harder to predict (Table S7). The pattern is the same as for the

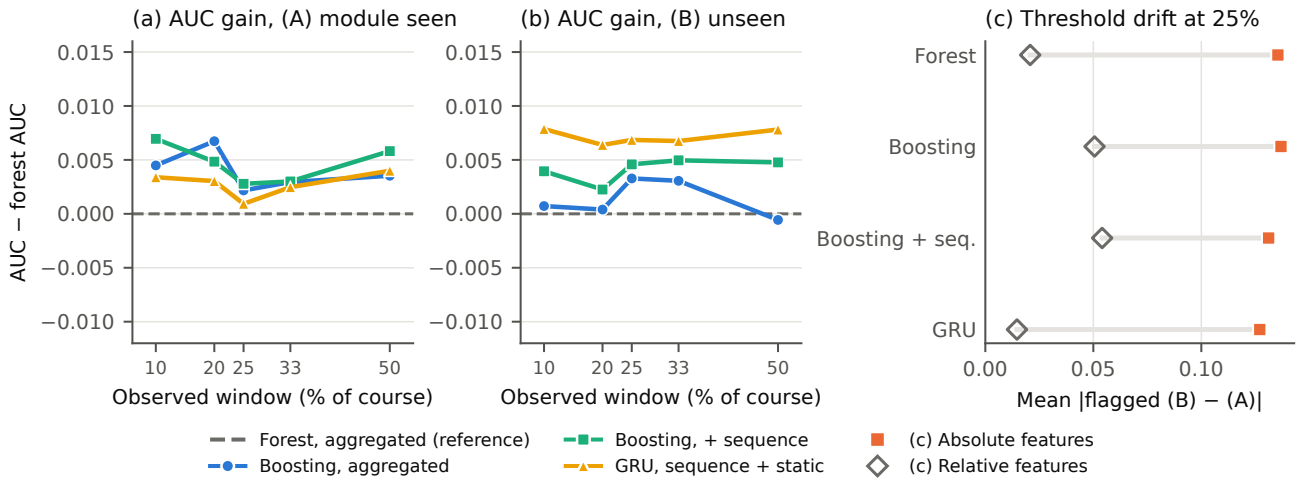


Figure S1: Sequence models and other learners. (a, b) AUC gain over the random forest on aggregated features (dashed line at zero) with relative features, when the module is seen (A) or unseen (B). (c) Mean absolute flag-rate drift at the 25% cut-off, with absolute and relative features.

Table S7: Robustness to the target definition (25% cut-off, students still enrolled; means over modules). W or F: Withdrawn or Fail; W only: Withdrawn only. Abs./rel.: absolute or relative features. Cost: AUC (A) − AUC (B); Drift: mean absolute flag-rate drift.

Target	Feat.	Risk (%)	AUC (A)	AUC (B)	Cost	Drift	GGG
W or F	abs.	38.7	0.800	0.787	+0.014	0.135	+0.51
W or F	rel.	38.7	0.816	0.808	+0.008	0.021	+0.00
W only	abs.	13.7	0.651	0.623	+0.028	0.054	+0.17
W only	rel.	13.7	0.675	0.662	+0.012	0.028	+0.03

main target. Relative features lower the ranking cost from 0.028 to 0.012 and the mean absolute drift from 0.054 to 0.028, and they raise the AUC on unseen modules from 0.623 to 0.662.

S8 Further robustness checks

Threshold. With absolute features, the mean absolute drift stays between 0.084 and 0.138 for every threshold from 0.3 to 0.7, and the drift of GGG between +0.16 and +0.60. With relative features, the mean absolute drift stays between 0.017 and 0.032.

Seeds. Across 10 seeds, which change both the forest and the assignment of folds, the standard deviation of a module's AUC when it is unseen is at most 0.004, and that of its drift at most 0.010.

Standard versus strict hold-out. If the students who are also enrolled in the held-out module are kept in the training modules, the mean AUC on unseen modules changes by at most 0.003. In the modules that share many students, removing the shared students changes the AUC of CCC by −0.007 and that of EEE by −0.006, whereas removing the same number of randomly chosen enrolments changes it by +0.001 and −0.001 respectively.

Training size. The training set of (B) is 0.93 to 1.22 times the size of that of (A). When the model that has seen the module is trained on exactly as many students as in (B), the transfer cost is 0.014 with absolute features and 0.008 with relative features, in line with the main analysis.

Table S8: Calibration of the transferred model at the 25% cut-off: difference between protocol (B) and protocol (A) on the same students, as the mean absolute difference over modules and for module GGG. Brier score; calibration-in-the-large (intercept of a logistic recalibration with logit($\hat{p}$) as offset); mean predicted risk; calibration slope. With balanced class weights the probabilities are not calibrated even in (A), so the transfer effect is the difference (B) − (A).

	Brier		Calibration-in-the-large		Mean predicted risk		Slope	
Features	mean $\lvert\Delta\rvert$	GGG	mean $\lvert\Delta\rvert$	GGG	mean $\lvert\Delta\rvert$	GGG	mean $\lvert\Delta\rvert$	GGG
Absolute	0.039	+0.180	0.419	−1.834	0.075	+0.298	0.170	−0.321
Relative	0.006	+0.004	0.079	−0.061	0.015	+0.011	0.077	−0.094

Table S9: Which cohort normalization? The eight engagement and early-assessment features are normalized within each presentation (weighted forest, 25% cut-off, LOMO strict, means over modules). $|\Delta^{\text{same}}|$: mean absolute flag-rate drift (B) − (A) of the same variant; |flagged − risk|: distance between the share flagged in (B) and the observed at-risk rate.

Normalization	AUC (A)	AUC (B)	Cost	$\lvert\Delta^{\text{same}}\rvert$	|flagged − risk|	Precision (B)	F1 (B)
Absolute values	0.800	0.787	+0.014	0.135	0.174	0.618	0.637
Percentile rank	0.816	0.808	+0.008	0.021	0.057	0.680	0.658
z-score (winsorized 1–99%)	0.814	0.801	+0.014	0.037	0.061	0.666	0.653
Robust (median, IQR)	0.809	0.800	+0.009	0.024	0.054	0.664	0.648

S9 Calibration

Table S8 gives the calibration measures summarized in the article. They compare the transferred model with the model that has seen the module, on the same students, so they measure how much calibration degrades under transfer.

S10 Choice of normalization

Table S9 compares percentile ranks with a winsorized z-score and a robust standardization. All three normalize within the cohort, and they give similar results.